WORKED SOLUTION

STEP III 1993, Question 14

Mechanics • Vertical Circular Motion

STEP III 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

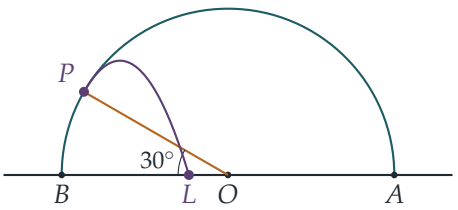
step.maths.org/questions/93-s3-q14

The key idea

Because the string is horizontal and the particle is projected vertically, the velocity starts perpendicular to the string, so the particle moves on a vertical circle of radius c about O, and it lands at the point B diametrically opposite A. Each impact is with the velocity vertical, so it simply halves the speed. We need to find the first bounce at which the particle can no longer get over the top of the circle, and then use energy and the radial equation to find where the tension vanishes. After that it is an ordinary projectile.

Solution

We model the particle as a particle of mass m, the string as light and inextensible, and the table as smooth and horizontal. Take O as origin, with $A = (c, 0)$ and $B = (-c, 0)$. Let the particle be at angle θ above the horizontal radius on whichever side it is rising.



Motion between impacts

Suppose the particle leaves the table at A or B moving vertically upwards with speed V. Its velocity is perpendicular to the taut string, so while the string stays taut it moves on the circle of radius c about O. By conservation of energy, at angle θ its speed v satisfies

$$v^2 = V^2 - 2gc\sin\theta.$$

Resolving towards O, with tension T,

$$T + mg\sin\theta = \frac{mv^2}{c} \quad \Longrightarrow \quad \frac{T}{m} = \frac{V^2}{c} - 3g\sin\theta.$$

For $0 \le \theta \le 90^\circ$ the tension is least at the top, so the string stays taut over the whole semicircle exactly when $V^2 \ge 3gc$. (Past the top, $\sin\theta$ decreases again, so T only grows.) In that case the particle comes down to the table at the opposite end of the diameter, moving vertically downwards with speed V again, by energy.

At that impact the velocity is normal to the table, so the rebound is vertical with speed $\frac{1}{2}V$, and the string is still horizontal and taut. So the whole process repeats with V halved.

Where the string goes slack

The particle starts with $V_0 = 64\sqrt{6gc}$, so $V_0^2 = 24576\,gc$. After k impacts the speed is $V_k = V_0/2^k$, so

$$V_k^2 = \frac{24576}{4^k}\,gc: \qquad V_6^2 = 6gc, \qquad V_7^2 = \tfrac{3}{2}gc.$$

So the semicircles starting with speeds $V_0, \ldots, V_6$ are all completed (each has $V_k^2 \ge 3gc$), giving impacts at B, A, B, A, B, A, B. The seventh impact is at B, and the particle rises from B with $V_7^2 = \frac{3}{2}gc < 3gc$.

> **Ask yourself** Can the particle fall back before the string goes slack, rather than the string going slack first?

No. The speed would vanish where $V_7^2 = 2gc\sin\theta$, that is $\sin\theta = \frac{3}{4}$, whereas the tension vanishes earlier, where

$$\frac{V_7^2}{c} = 3g\sin\theta \quad \Longrightarrow \quad \tfrac{3}{2}gc = 3gc\sin\theta \quad \Longrightarrow \quad \sin\theta = \tfrac{1}{2}.$$

So the string becomes slack at $\theta = 30^\circ$, at the point P on the B side, at height $c\sin\theta = c/2$ above the table, as required.

The first rebound between A and B

At P we have $\theta = 30^\circ$, so $P = \left(-\frac{\sqrt{3}}{2}c,\ \frac{1}{2}c\right)$ and

$$v^2 = \tfrac{3}{2}gc - 2gc\cdot\tfrac{1}{2} = \tfrac{1}{2}gc.$$

The velocity is along the tangent, perpendicular to OP, pointing up and towards O; it makes 60° with the horizontal. So its components are

$$v_x = \tfrac{1}{2}v, \qquad v_y = \tfrac{\sqrt{3}}{2}v, \qquad \text{with } v_x^2 = \tfrac{1}{8}gc, \quad v_y^2 = \tfrac{3}{8}gc.$$

From now on the particle is a projectile. It reaches the table when $\frac{1}{2}c + v_y t - \frac{1}{2}gt^2 = 0$; taking the positive root,

$$t = \frac{v_y + \sqrt{v_y^2 + gc}}{g} = \frac{\sqrt{3}+\sqrt{11}}{\sqrt{8}}\sqrt{\frac{c}{g}}.$$

Its horizontal displacement in this time is

$$v_x t = \sqrt{\frac{gc}{8}} \cdot \frac{\sqrt{3}+\sqrt{11}}{\sqrt{8}}\sqrt{\frac{c}{g}} = \frac{\sqrt{3}+\sqrt{11}}{8}\,c,$$

so it lands at the point L with

$$x_L = -\frac{\sqrt{3}}{2}c + \frac{\sqrt{3}+\sqrt{11}}{8}c = \frac{\sqrt{11}-3\sqrt{3}}{8}\,c \approx -0.235\,c.$$

(Throughout this flight the particle stays within distance c of O, so the string does not become taut again before L; in particular $|x_L| < c$.)

Since $\sqrt{11} < \sqrt{27} = 3\sqrt{3}$, we have $x_L < 0$: the landing point lies on the same side of O as B. As O is the midpoint of AB, the first rebound between A and B is **nearer to** B, at distance about $0.77c$ from B and $1.23c$ from A.

Checks

- The test $V_k^2 \geq 3gc$ picks out $k = 7$ as the first incomplete semicircle, and $V_7^2 = \frac{3}{2}gc$; the slack condition gives $\sin\theta = \frac{1}{2}$.
- At P the radial component of gravity, $g\sin 30^\circ = \frac{1}{2}g$, equals $v^2/c = \frac{1}{2}g$, just as zero tension requires.
- A numerical step-by-step simulation (with $g = 9.8$, $c = 1.3$) found the string going slack at height $0.5000c$, a landing point at $x \approx -0.2349c$, and a distance from O never exceeding c during the flight.

Common slips

- Using $V^2 \geq 4gc$ (just reaching the top) as the condition for completing the semicircle. A string, unlike a rod, needs $V^2 \geq 3gc$.
- Losing track of which end the seventh impact is at: odd impacts are at B.
- Giving the velocity at P the wrong direction: it is perpendicular to OP, at 60° to the horizontal, not 30°.

Beyond STEP

The moment when the string goes slack is a constraint switching off: the tension is the Lagrange multiplier that enforces $|OP| = c$, and when it would have to become

negative the constraint stops acting. Problems with one-sided constraints like this, and with impacts, are treated systematically in undergraduate mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see energy and radial equations used together in this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.